

Jensen's terms arise because of the difference between *geometric* returns, which take into account the compounding over the long run, and *arithmetic* returns, which do not compound.⁴⁰ In a one-period setting, geometric and arithmetic returns are economically identical; they are simply different ways of reporting increases or decreases in wealth. Thus, there is no rebalancing premium for a short-run investor. Over multiple periods, the difference between geometric and arithmetic returns is a function of asset volatility, specifically approximately $\frac{1}{2}\sigma^2$, where σ is the volatility of arithmetic returns. The greater the volatility, the greater the rebalancing premium. As this manifests over time, only long-term investors can collect a rebalancing premium.

For U.S. stocks, the rebalancing premium a long-run investor can earn is approximately $\frac{1}{2}(0.2)^2 \approx 1\%$. Erb and Harvey (2006) estimate a rebalancing premium of around 3.5% in commodities. These are significant premiums for simple, automatic rebalancing. In his 2009 book, David Swensen, the superstar manager of Yale University's endowment, emphasizes that rebalancing plays an important role in his practice of investment management, especially in the daily rebalancing of Yale's liquid portfolio. He refers to a "rebalancing bonus" arising from maintaining a constant risk profile.

So, is rebalancing optimal not only because it reduces risk, but also because it provides a "free lunch" in the form of a rebalancing premium? Not so fast. In section 3, I showed that rebalancing has no value in the market by interpreting rebalancing as an option strategy. The rebalancing premium seems too good to be true—and in fact it is. Rebalancing is a short volatility strategy that does badly compared to buy and hold when asset prices permanently continue exploding to stratospheric levels or permanently implode to zero and disappear. Rebalancing is short a regime change. The crucial assumption behind the rebalancing premium is that the assets over which you rebalance continue to exist. If there are assets that experience total irreversible capital destruction, then rebalancing leads to buying more assets that eventually disappear—this is wealth destruction, not wealth creation. The rebalancing premium can only be collected for assets that will be around in the long run, so rebalance over very broad asset classes or strategies: global equities, global sovereign bonds, global corporate bonds, real estate, commodities, and so on, rather than individual stocks or even individual countries.

5.2. THE VERY LONG RUN

In the very long run, the portfolio that maximizes wealth is a rebalanced portfolio that holds constant asset weights that maximize the rebalancing premium. This

⁴⁰The arithmetic return r represents $(1+r)$ at the end of the period. The same amount can be expressed as a geometric return, g , where $(1+r) = \exp(g)$. The means of the arithmetic return and the geometric return are related by $E(r) \approx E(g) - \frac{1}{2}\sigma^2$, where σ is the volatility of r . This relation holds exactly for log-normal distributions. For more details, see the appendix.